Zhenye Luo

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EDUCATION & EXPERIENCES

Mohamed bin Zayed University of Artificial intelligence (MBZUAI), Abu Dhabi, UAE 2025.08 – Present
PhD Candidate in Robotics

Westlake University (Westlake), Hangzhou, ZJ, China 2024.11 – 2025.07
Research Assistant in Artificial Intelligence and Robotics **Advisor: Prof. Yaochu Jin**

Beijing Normal University (BNU), Beijing, BJ, China 2021.09 – 2024.07
Master of Science in Telecommunications and Information Systems GPA:3.86/4.0 **Advisor: Prof. Li Yao**

Beijing University of Chemical Technology (BUCT), Beijing, BJ, China 2017.09 – 2021.07
Bachelor of Automation GPA: 3.94/4.33

PUBLICATIONS & MANUSCRIPTS

[1] **Luo Z**, *Ren M, Hu X, et al.* Popdg: Popular 3d dance generation with popdanceset[C]//Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2024: 26984-26993.
[paper] [project page]

PROFESSIONAL SERVICE

Reviewer, International Journal of Computer Vision (IJCV) 2025–present
Reviewer, NeurIPS 2025–present

RESEARCH EXPERIENCE

Trustworthy and General Artificial Intelligence Laboratory, Westlake **Advisor: Yaochu Jin**
Research Assistant 2024.11 – Present

Research on Music-Driven Conducting Robots Based on Diffusion Models

- Built a 3D motion dataset for music conducting, featuring videos from renowned conductors and teaching instructors.
- Developed a diffusion-based generation framework to create conducting motion sequences utilizing multi-modal data.
- Applied optimization algorithms to retarget the smpl-based music conducting motion to the humanoid robot (Unitree G1).

State Key Laboratory of Cognitive Neuroscience and Learning, BNU **Advisor: Prof. Li Yao**
Master's Graduation Thesis 2022.06 – 2024.06

3D Dance Generation Driven by Music Data Based on Improved-DDPM

- Developed a dance popularity function using multivariate linear regression based on a data set of dance videos that was created over a year.
- Constructed the PopDanceSet music-dance dataset, utilizing monocular 3D pose estimation techniques.
- Designed a framework for the generation of music-driven dance using the improved DDPM (iDDPM).
- Innovated the Space Augmentation Algorithm and Alignment Module to enhance spatial relationships between human joints, improving the match between generated dances and music.
- Proposed the new Physical Body Contact (PBC) metric as an innovative evaluation metric.

Microfluidic Chip Biomedical Engineering Research Center, BUCT **Advisor: Prof. Xianbo Qiu**
Undergraduate Research Assistant 2019.11 – 2021.01

Centrifugal Microfluidic Chip Technology and Its Applications

- Fluid physics force analysis during high-speed rotation to derive optimal dimensions (radius, width, and depth) for microfluidic chip channels.
- Designed comprehensive 3D models of microfluidic chips using SolidWorks software.
- Fabricated centrifugal microfluidic chips on polymethylmethacrylate (PMMA) using laser cutting and CNC milling techniques.

SKILLS & EXPERTISE

- **Mathematical and Control Theory Proficiency:** Solid foundation in Mathematics and Physics, skilled in both classical and modern control theories.
- **Programming and Simulation:** Proficient in Python, PyTorch, and MATLAB, familiar with C++, and adept at building simulation models using SIMULINK.
- **AI and Robotics:** Proficient in using the Ubuntu system, with a solid understanding of deep learning models in computer vision and AIGC, as well as hands-on experience with robotics simulation platforms such as Isaac Sim and Mujoco.
- **Mechanical Design and Experimental Operation:** Known for using mechanical drawing and simulation software such as SolidWorks and experienced in operating CNC milling machines and laser cutters.
- **English Expertise:** IELTS: Overall: 7.5, Listening: 8.5, Reading: 9.0, Writing: 7.0, Speaking: 6.0.

HONORS & AWARDS

Excellent Graduate of Beijing (Top 4%)	2024
Excellent Graduate of Beijing Normal University (Top 10%)	2024
First Class Academic Scholarship (Academic Master's Degree), *BNU*	2022 and 2023
Outstanding Undergraduate Teaching Assistant for the 2022-2023 Academic Year, *BNU*	2023
Bronze Medalist in the 19th Graduate Football League, BNU	2023
Eighth Place in The 3rd International Competition on Human Identification at a Distance 2022	2022
First Prize in the 11th Humanities Knowledge Competition, *BNU*	2022
Finalist in the American Mathematical Contest in Modeling (**Globally Top 1.3%**)	2020
Third Prize in the 11th National College Student Mathematics Competition (Non-math Major)	2019
First Prize in the 10th Beijing College Student Humanities Knowledge Competition (Team), 3rd place	2019
Second Prize in the 10th Beijing College Student Humanities Knowledge Competition (Individual)	2019
Second Prize in Beijing Division, Group A, National College Student Mathematical Modeling Contest	2018
Second Prize in the Zhongkong Scholarship, *BUCT*	2018
Outstanding Student of BUCT	2018
Excellent Student of BUCT (Top 5%)	2017

EXTRACURRICULAR ACTIVITIES

Undergraduate Teaching Assistant for C++ Class, Spring Semester 2022-2023, BNU 2023.02 – 2023.07

- Facilitated learning and provided support to undergraduates in the C++ programming class (79 students).

Academic Affairs Director, Graduate Student Union, School of AI, BNU 2021.10 – 2022.10

- Led the organization and planning of academic salons and doctoral forums, including inviting speakers and hosting events, contributing to the Student Union being recognized as outstanding for the academic year.

Volunteer for the "Siemens Cup" China Intelligent Manufacturing Challenge 2018.07 – 2018.07

- Supported the organization and execution of the competition as a volunteer.

Debater for the College of Information Science and Technology, BUCT 2017.10 – 2018.10

- Served as a debater for the college's debate team, engaging in intellectual and competitive debates.

Member of the Science and Technology Innovation Department, Student Union, BUCT 2017.10 – 2018.10

- Planned and organized various science and technology innovation activities at BUCT, including serving as the lead organizer of the Encyclopedia Contest (School level).